{EPITECH}

BOOTCAMP DATA_

< RUSH 3 - CUSTOMER SATISFACTION />



BOOTCAMP DATA

Introduction

More than half of an airline's passengers describe their flight as neutral or disappointing. None of them is lost yet, but none of them is a reason to book again either. The airline has surveyed over a hundred thousand passengers and wants two things from that survey: to know what actually drives satisfaction, and to spot unhappy passengers early enough to do something about it.

Your role

Context

You are a Data / Business Intelligence consultant working for a European airline. The Customer Experience director has a budget to improve the passenger journey and wants to spend it where it matters. They also want the service teams to be able to approach passengers who are likely to be dissatisfied before they leave the arrival airport.

The director is not a data scientist. They will fund what they understand, and distrust what they cannot explain to their own executive committee.

Objective

Identify the factors that explain passenger satisfaction, build a classification model that flags passengers likely to be dissatisfied at the moment the service teams can still act, and turn both into recommendations the director can act on.

Logistics

Item	Detail
Group	3 people
Duration	2 days
Tools	Python or R
Data	Survey results, split in two files

Expectations

The data

Each row is one passenger's answers to the post-flight survey, together with their profile and flight information. Service ratings go from 1 to 5; a 0 means the passenger did not answer or the service did not apply to them. The survey classifies each passenger as either ”satisfied” or ”neutral or dissatisfied”. Respondents of all ages took part, children included.

The first file is yours to explore, clean and model. The second one stands for next month's passengers: you have it, but the airline will judge your model as if you did not. It may only be used once, at the very end, to report the performance of the model you chose, and your code must make that visible.



Using the second file to explore the data, select variables or tune the model makes your reported performance meaningless. The airline would be making decisions on a number that does not reflect reality.



Since the second file cannot help you choose between models, you will need another way to compare them.

Explaining satisfaction

The director will allocate budget to the factors you name. A factor that is statistically associated with satisfaction is not necessarily one that the airline can act on, nor one that causes satisfaction. Your report must make the difference explicit for each factor you recommend.

Predicting dissatisfaction

The service teams act when the passenger lands, not when the survey comes back. Your model may only use what the airline knows at that moment, and it is up to you to establish what that is. It will be trained on passengers who answered the survey, and applied to every arriving passenger.

You must compare at least two different classification approaches, and show how much better they do than the simplest rule the service teams could apply without any model. The choice of the model you would deploy must go beyond the highest score: the director needs to understand what the model does and what its errors would cost. Their own estimates:

- ✓ approaching a passenger costs the airline about €40;
- ✓ about one neutral or dissatisfied passenger in three does not fly with the airline again, which represents about €600 of lost future revenue;
- ✓ an approach brings back about half of those who would otherwise leave;
- ✓ the service teams can approach at most one arriving passenger in ten.

Data protection

The survey was collected to measure satisfaction. Flagging arriving passengers to airport staff — most of whom never answered any survey — is a new use of their data. Your report must state the conditions under which it would be acceptable, including for children, whether each piece of information your model uses is acceptable for that purpose, and what the service teams would actually be shown about a flagged passenger.

Presentation

At the end of the Rush, your group presents its conclusions to the director (played by a staff member) in **10 minutes**, followed by questions on your analytical choices.

Deliverables

Deliverable	Content
Report (PDF, 8 pages max, excl. annexes)	Opens with a one-page summary for the director; exploration, factors, models, recommendations, data protection
Code	Notebook or scripts reproducing every figure and result in the report
Oral presentation	10 minutes, followed by questions



Every figure quoted in the report must be reproducible by running the delivered code as is, and running it twice must give the same figures.

The group repository is part of the deliverables: its history must show how the work was shared, and its README must state who was responsible for what.

Resources

The two survey files are available in the student resources of the module (`rush-3/`).



v 2.0.1

